Last time:

- For positive integers m, n, the **Ramsey number** $R(m,n)$ is the smallest integer N such that every red,blue-coloring of $E(K_N)$ contains either a red copy of K_m or a blue copy of K_n.
- To prove that $R(m,n) \geqslant N$, we must **construct a** red,blue-coloring of $E(K_{N-1})$ which has neither a red copy of K_m nor a blue copy of K_n.
- To prove that $R(m,n) \leqslant N$, we must **show that every** red,blue- coloring of $E(K_N)$ contains either a red copy of K_m or a blue copy of K_n.
- $R(n,m)$ exists for any n, m.
- $R(3,3) = 6$.
- $R(4,3) = 9$.
- $R(n,n) \leqslant 4^n$.

$$R(m,n) \leq R(m-1,n) + R(m,n-1).$$

Theorem. $R(4,4) = 18$.

Proof: $R(4,4) \leq 18$: $R(4,4) \leq R(3,4) + R(4,3) = 18$

$R(4,4) \geq 18$ Need to construct $f: E(K_{17}) \longrightarrow \{r, b\}$

with no monochromatic K_4.

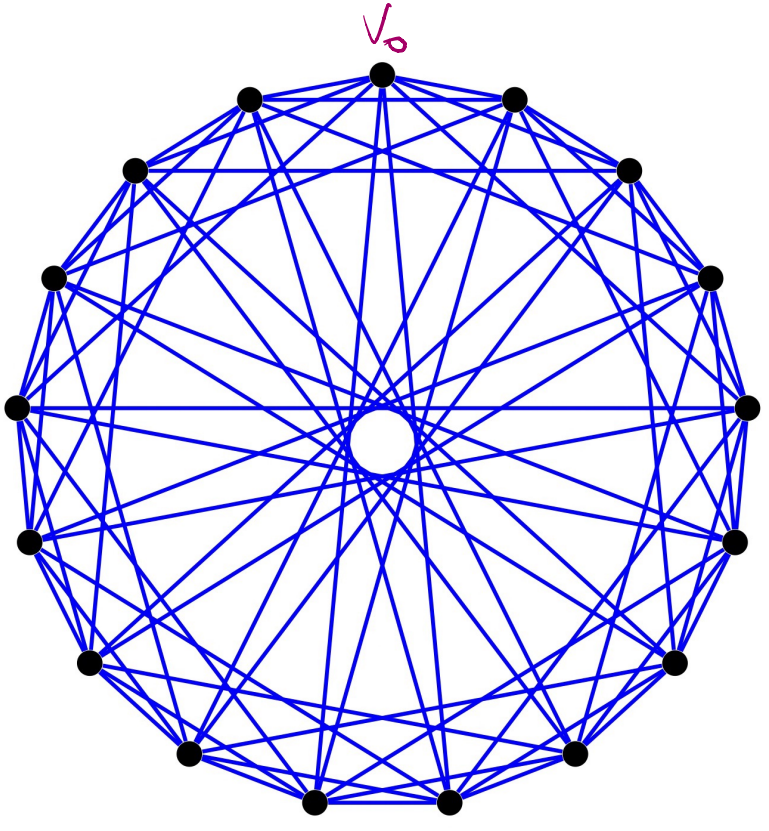
Label $V(K_{17}) = \{v_0, v_1, \ldots, v_{16}\}$ and define

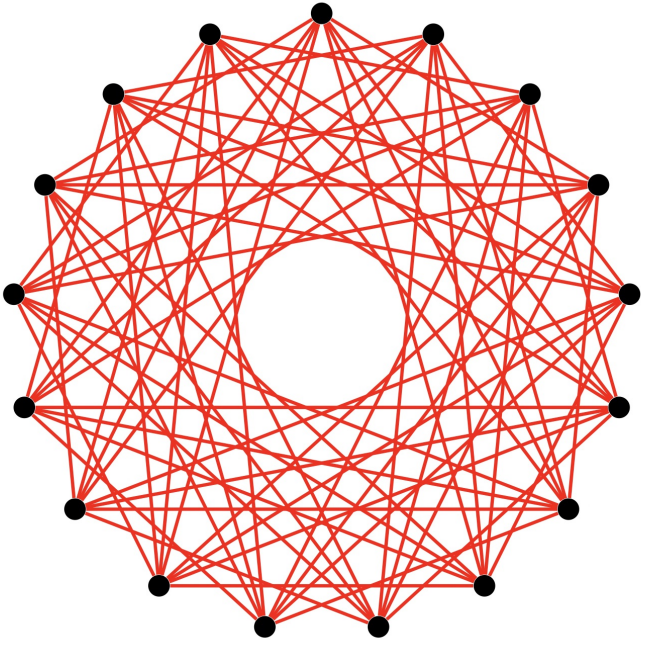
$f(v_i v_j) = b$ if $i - j \in \{\pm 1, \pm 2, \pm 4, \pm 8\}$ mod 17.

These are all the squares mod 17

$f(v_i v_j) = r$ otherwise "Paley Graph"

One can see(!) There is no monochromatic K_4





Courtesy Chris Cox.

$R(4,4) = 18$

Best available bounds:

$$43 \leqslant R(5,5) \leqslant 48, \qquad 102 \leqslant R(6,6) \leqslant 165.$$

$R(n,n) \leq 4^n$, no bound $R(n,n) \leq (4-\varepsilon)^n$

Theorem. $R(n,n) \geqslant \sqrt{2}^n$ *for any* $n \geqslant 2$,

$R(n,n) \geq (\sqrt{2})^n$

Proof: probability.

No bound $R(n,n) \geq (\sqrt{2}+\varepsilon)^n$ is known.

Definition. *Given k positive integers $n_1, \dots, n_k$, the **Ramsey number** $R_k(n_1, \dots, n_k)$ is the minimum N such that any coloring of $E(K_N)$ using k colors contains a clique of size n_i in color i, for some i.*

Theorem. *For any positive integers $n_1, \dots, n_k$, we have*

$$R_k(n_1, \dots, n_{k-2}, n_{k-1}, n_k) \leqslant R_{k-1}(n_1, \dots, n_{k-2}, R(n_{k-1}, n_k)).$$

In particular, the Ramsey number $R_k(n_1, \dots, n_k)$ exists.

Proof: For existence, induct on k.

For the $\leq$, let $N = R_{k-1}(n_1, \dots, n_{k-2}, R(n_{k-1}, n_k))$

Take $f: E(K_N) \longrightarrow \{1, \dots, k\}$ any coloring.

WTS: f contains a K_{n_i} in color i, for some $i \in \{1, \dots, k\}$.

Regard $k-1, k$ as the same "blurred color". Thus we have a

$\tilde{f}: E(K_N) \longrightarrow \{1, \dots, k-2, \{k-1, k\}\}$ $(k-1)$-coloring of $E(K_N)$

Thus, by definition of N, either

1) $\tilde{f}$ contains a K_{n_i} in color i for some $i \in \{1, \ldots, K-2\}$

2) $\tilde{f}$ contains a $K_{R(n_{K-1}, n_K)}$ in blurred color $\{K-1, K\}$

In case (1) we are done

In case (2) we recover sight, and see that we have a 2-coloring of $K_{R(n_{K-1}, n_K)}$, using colors $K-1, K$.

By definition of $R(n_{K-1}, n_K)$, this last coloring contains either $K_{n_{K-1}}$ in color $K-1$ or a K_{n_K} in color n_K

so we are also done $\blacksquare$

$R(3,3,3) \le R(3, R(3,3)) = R(3,6)$

Theorem. $R(3,3,3) = 17$.

Proof: $R(3,3,3) \le 17$ HW 13.2.

Fix any $f: E(K_{17}) \longrightarrow \{r,b,g\}$. Fix any $v \in V(K_{17})$.

$N(v) = R \cup B \cup G$, $|N(v)| = 16$

Wlog $|R| \ge 6 = R(3,3)$

↓ Hint.

$R(3,3,3) \ge 17$: Enough to construct a coloring

$f: E(K_{16}) \longrightarrow \{r,b,g\}$

with no monochromatic Triangle

Idea (for blue edges)

Since $16 = 2^4$, there is a field $\mathbb{F}_{16}$ with 16 elements

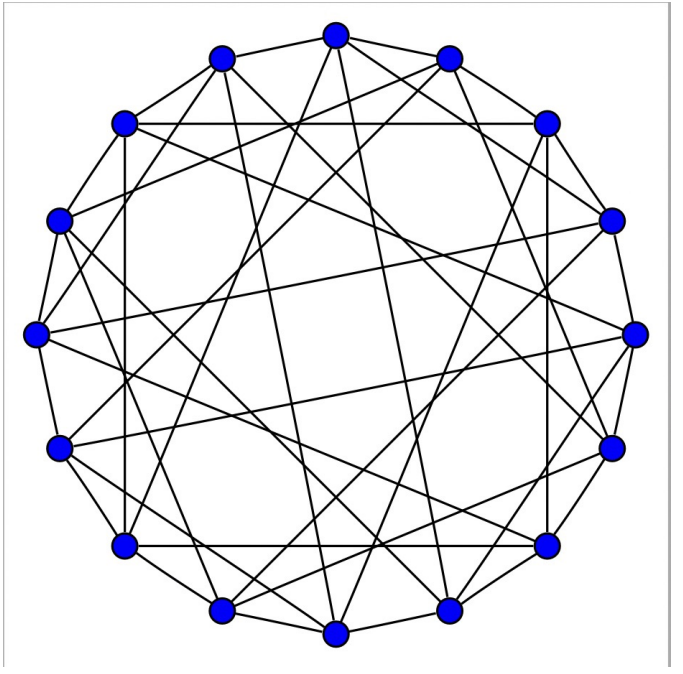
Label $V(K_{16}) = \{v_x : x \in \mathbb{F}_{16}\}$

Define $f : E(K_{16}) \longrightarrow \{r, g, b\}$

$f(v_x v_y) = b$ if $x - y$ is a cube in $\mathbb{F}_{16}$

Similar for red and green

One can see there is no monochromatic triangle.



Courtesy Wikipedia.

∎

Theorem (Schur, 1916). *For any* [redacted], *there is* $N > 3$ *such that for any* k*-coloring of* $\{1, 2, \ldots, N\}$, *there are three integers* x, y, z *of the same color such that* $x + y = z$.

Proof: $N = R_k(\underbrace{3, \ldots, 3}_{k})$. By definition, any coloring of $E(K_N)$ using k-colors will contain a monochromatic Triangle.

Given any coloring $c: \{1, \ldots, N\} \longrightarrow \{1, \ldots, k\}$, we define

$f: E(K_N) \longrightarrow \{1, \ldots, k\}$, label $V(K_N) = \{v_1, \ldots, v_N\}$

$f(v_i v_j) = c(j - i \mod N)$. By definition of N, f has a monochromatic Triangle, say $\{v_i, v_j, v_k\}$ where $i < j < k$. Define

$x = j - i$, $y = k - j$, $z = k - i \in \{1, \ldots, N\}$. Now

$c(x) = c(y) = c(z)$ and $x + y = (j - i) + (k - j) = k - i = z$ ∎

Theorem. *For each* $m \geq 1$, *there is* p_0 *such that for any prime* $p \geq p_0$, *the congruence*

$$x^m + y^m \equiv z^m \mod p$$

has a solution

Fermat's Theorem is false in any $\mathbb{Z}_p$ for big p

"Proof": There is an integer g such that any integer x not divisible by p is of the form

$$x = g^n \mod p \quad \text{for some } n$$

Write $n = mq + i$, $i \in \{0, \dots, m-1\}$, define

$$C(x) = i \quad \text{if} \quad x = g^{mq+i}$$

Apply Previous Thm . . . ∎